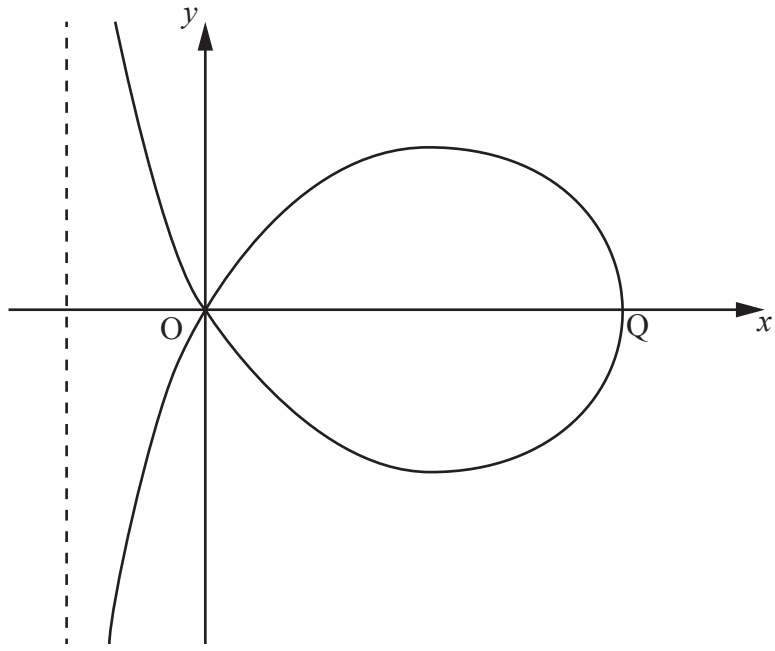
14 **(a)** Draw and label a suitable triangle on the diagram in the **Printed Answer Booklet** to show that $\tan\theta = \frac{y}{x}$, as given in line 30. **[1]**

(b) Subtract $x = \frac{a(3-t^2)}{1+t^2}$ from $3a$ to show that $3a - x = \frac{4at^2}{1+t^2}$. **[1]**

(c) Find an expression, in terms of a and t, for $a+x$. **[1]**

(d) Hence, or otherwise, show that the parametric equations given in lines 29 and 31 are equivalent to $y^2 = \frac{x^2(3a-x)}{a+x}$, as claimed in lines 8 and 33. **[2]**

14(a)



14(b)

14(c)

14(d)